

House Price Prediction System

AI Pioneers ML – Week 4

End-to-End Machine Learning Project

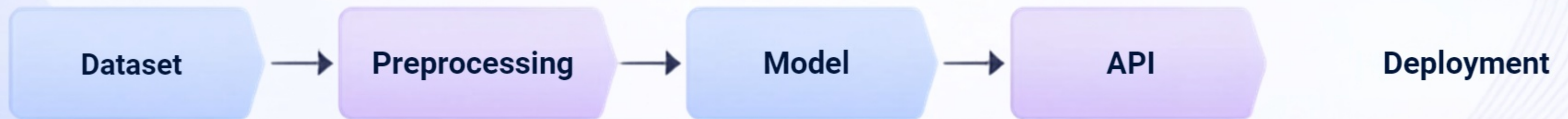
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Project Objective



Dataset

California Housing Dataset

20,640

Samples

8

Input Features

80:20

Train-Test Split

HouseValue

Target Variable

Features Used

MedInc — Median income

HouseAge — Median house age

AveRooms — Average number of rooms

AveBedrms — Average number of bedrooms

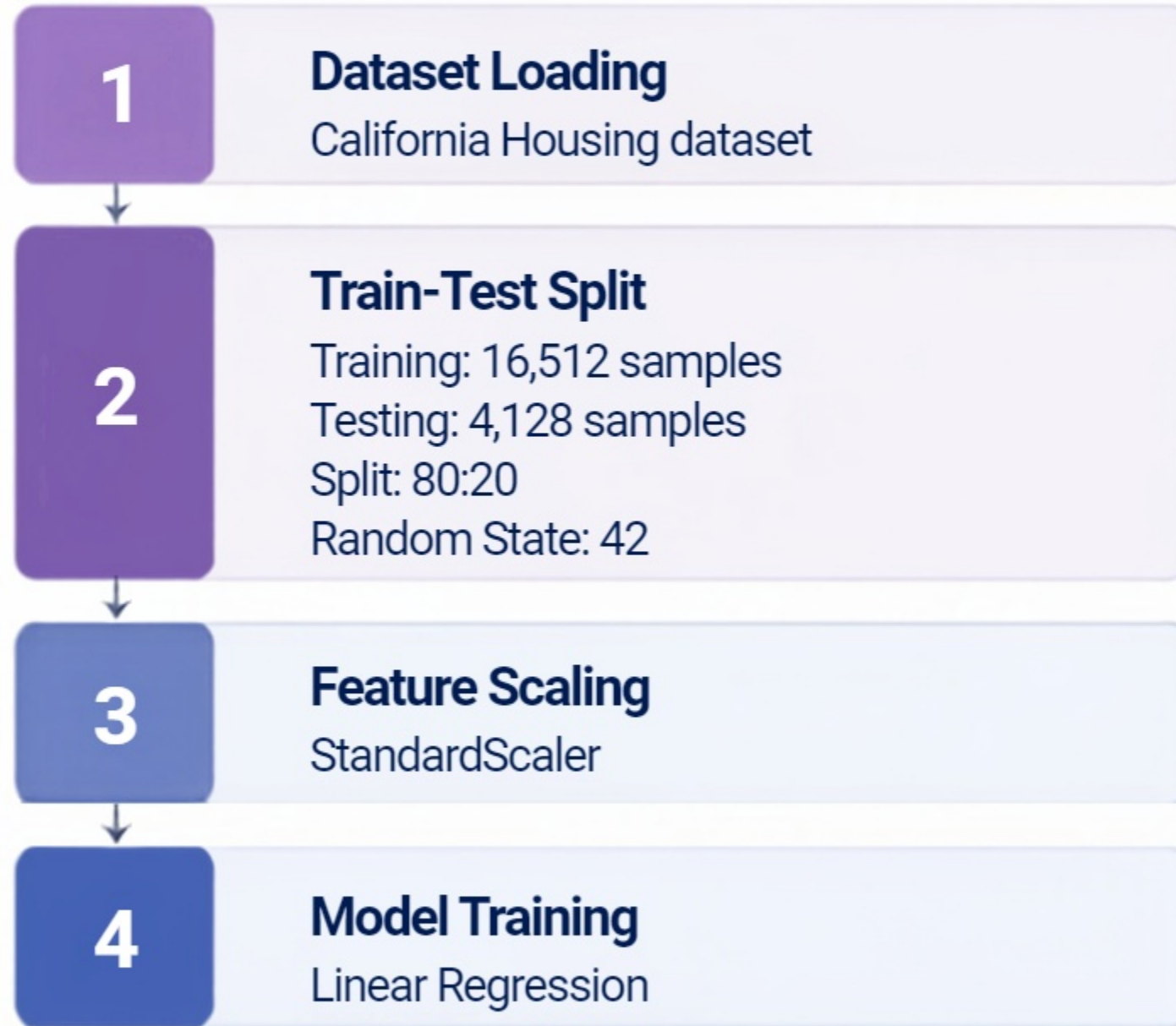
Population — Block population

AveOccup — Average house occupancy

Latitude — Latitude of the location

Latitude — Latitude of the location

Data Preprocessing and Model Training



StandardScaler was used to standardize the numerical input features before training the LinearRegression model.



Model Performance

MAE

0.5332

MSE

0.5559

RMSE

0.7456

R² Score

0.5758

R² Score: 57.58%

The model explains approximately 57.58% of the variation in house values in the test data.

The project focuses on demonstrating an end-to-end Machine Learning and deployment workflow rather than highly optimized prediction accuracy.

Model Serialization and Flask API

Model Serialization



Saved Model:
house_price_model.joblib

Saved Scaler:
house_price_scaler.joblib

The trained model and StandardScaler were saved using Joblib so they can be loaded later without retraining.

Flask Prediction API



GET /
Confirms that the API is running.

POST /predict
Accepts house-related input and returns a prediction.

Features accepted:

Medinc	HouseAge	AveRooms	AveBedrms
Population	AveOccup	Latitude	Longitude



Deployment and API Testing



DEPLOYMENT

Platform: Render

Build Command:

`pip install -r requirements.txt`

Start Command:

`gunicorn app:app`



API TEST RESULT

HTTP Status Code:

200

Predicted House Value:

2.4731

API Status:

Successfully tested



LIVE APPLICATION

Live URL:

<https://ai-pioneers-ml-week4.orienter.com>

“The deployed prediction endpoint was successfully tested and returned a prediction.”



GitHub



Render



Live Flask API

Conclusion and Learning Outcomes

Conclusion

The House Price Prediction System demonstrates a complete end-to-end Machine Learning workflow, from data preprocessing and model training to model serialization, API development, and cloud deployment.



Learning Outcomes

- Dataset preparation and preprocessing
- Train-test splitting
- Feature scaling using StandardScaler
- Linear Regression
- MAE, MSE, RMSE and R^2 evaluation
- Model serialization using Joblib
- Flask API development
- getJSON-based prediction requests and responses
- GitHub project management
- Cloud deployment using Render



GitHub Repository:

[https:// github.com/deepashreea10/Al-Pioneers-ML-Week4](https://github.com/deepashreea10/Al-Pioneers-ML-Week4)